Dehuller



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: De-huller; Dehulling Machine

What it is used for

A dehuller removes the outer husk or hull from suitable grains, pulses, or seeds using a controlled mechanical process. Correct crop preparation, machine adjustment, feed control, and safe separation of product and husk help improve processing quality and reduce waste and injury risks.

Main parts

- Feed hopper receives the crop material.
- Feed gate or metering device regulates material flow.
- Dehulling chamber contains the working mechanism.
- Abrasive surface, rollers, blades, or impact mechanism loosens the hull depending on the model.
- Drive motor and transmission power the working mechanism.
- Outlet discharges the dehulled product.
- Husk outlet or separator removes separated hulls where fitted.
- Screen or sieve separates product fractions where fitted.
- Guards and emergency stop protect the operator.

Before you start

1. Only trained personnel should operate the dehuller.
2. Read the manual and confirm that the machine is suitable for the crop type, moisture condition, and required output.
3. Inspect the hopper, chamber, drive, belts, rollers, screens, outlets, guards, and electrical connections for damage.
4. Ensure the machine is clean and free from residue or foreign material from previous batches.
5. Confirm that all guards are fitted, the emergency stop functions, and the machine is securely mounted.
6. Set the working clearance, speed, and feed rate according to the crop and manufacturer guidance.
7. Wear suitable eye protection, hearing protection, safety footwear, and respiratory protection where dust exposure is possible.

How to use

1. Prepare and inspect the crop, removing stones, metal, and other foreign objects using the approved pre-cleaning method.
2. Place suitable collection containers at the product and husk outlets.
3. Ensure the discharge paths are open and all covers and guards are secured.

4. Start the dust extraction system where fitted and switch on the machine according to the manual.
5. Allow the drive to reach its specified operating condition before feeding material.
6. Feed the crop steadily through the hopper without exceeding the rated capacity.
7. Observe the dehulling process, product flow, husk separation, noise, and vibration during operation.
8. Adjust feed rate or approved machine settings only when permitted and using the safe procedure.
9. Stop feeding before shutdown and allow material inside the chamber to clear as instructed.
10. Switch off and isolate power, then wait for all moving parts to stop completely.
11. Collect and inspect the dehulled product; check hull removal, broken kernels, and product quality against process requirements.
12. Clean the hopper, chamber, screens, and outlets after isolation and record production results or faults.

⚠ Safety

- Never reach into the hopper, chamber, or outlet while the machine is operating.
- Do not operate with missing guards, damaged belts, loose parts, or a faulty emergency stop.
- Isolate and lock out power before clearing blockages, changing screens, or performing maintenance.
- Control grain dust using suitable extraction and housekeeping; airborne dust may cause respiratory irritation and, in some conditions, a combustible dust hazard.
- Keep hands, hair, clothing, and tools away from rotating and moving components.
- Do not process wet, contaminated, or unsuitable material beyond the machine's approved limits.

Care & storage

- Remove crop residue and dust after each batch using an approved cleaning method.
- Inspect rollers, abrasive surfaces, screens, belts, bearings, and fasteners for wear or damage.
- Lubricate designated bearings and drive components according to the manufacturer's schedule.
- Keep the machine dry and protect electrical parts from dust and moisture.
- Arrange authorised maintenance if dehulling quality drops or abnormal noise, vibration, or overheating occurs.

Common mistakes to avoid

- Feeding material faster than the machine can process it.
- Using incorrect clearance or speed for the crop type.
- Allowing stones or metal to enter the dehulling chamber.
- Cleaning or clearing blockages before isolating power and confirming all parts have stopped.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.